EE 533 Information Theory

Week 08

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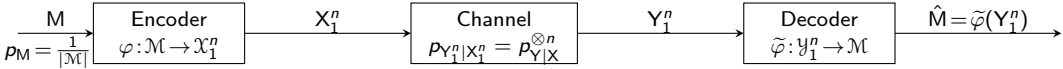
Channel Coding Problem and Channel Coding Theorem



DMC $p_{\mathsf{Y}_1^n|\mathsf{X}_1^n}(y_1^n|x_1^n) = \prod_{t=1}^n p_{\mathsf{Y}|\mathsf{X}}(y_t|x_t) \qquad \forall x_1^n, y_1^n$

Encoding function $\varphi : \mathcal{M} \to \mathcal{X}_1^n$

Decoding function $\widetilde{\varphi} : \mathcal{Y}_1^n \to \mathcal{M}.$

$$P_{\mathbf{e}}^{(n)} := \mathbf{P}\left(\hat{\mathsf{M}} \neq \mathsf{M}\right) \qquad P_{\mathbf{e}}^{(n),max} := \max_{m \in \mathcal{M}} \mathbf{P}\left(\hat{\mathsf{M}} \neq \mathsf{M} \middle| \mathsf{M} = m\right)$$

Theorem

For any DMC $p_{\mathsf{Y}|\mathsf{X}}$, all rates below the Shannon capacity of $p_{\mathsf{Y}|\mathsf{X}}$ are achievable: for every rate $R < C\left(p_{\mathsf{Y}|\mathsf{X}}\right)$, there exists a sequence of $(\lceil 2^{nR} \rceil, n)$ codes satisfying $\limsup_{n\to\infty} P_{\mathbf{e}}^{(n),max} = 0$. Conversely, for any sequence of $(\lceil 2^{nR} \rceil, n)$ codes on $p_{\mathsf{Y}|\mathsf{X}}$ with a $R > C\left(p_{\mathsf{Y}|\mathsf{X}}\right)$, the maximum probability of error is bounded away from zero, i.e. $\liminf_{n\to\infty} P_{\mathbf{e}}^{(n),max} > 0$.

Proof of the Weak Converse Part I: Fano's Inequality

$$H\left(\mathsf{M}\middle|\hat{\mathsf{M}}\right) \leq \hbar\left(P_{\mathsf{e}}^{(n)}\right) + P_{\mathsf{e}}^{(n)}\log(\lceil 2^{nR}\rceil - 1) \qquad \text{by Fano's inequality}$$

$$\leq 1 + P_{\mathsf{e}}^{(n)} nR$$

Proof of Fano's Inequality.

$$\begin{aligned}
H\left(\mathsf{M}\middle|\hat{\mathsf{M}}\right) &= H\left(\mathsf{M}\middle|\hat{\mathsf{M}}\right) + H\left(\mathbb{1}_{\{\hat{\mathsf{M}}\neq\mathsf{M}\}}\middle|\hat{\mathsf{M}},\mathsf{M}\right) && \text{because } H(f(\mathsf{Z})|\mathsf{Z}) = 0 \\
&= H\left(\mathsf{M},\mathbb{1}_{\{\hat{\mathsf{M}}\neq\mathsf{M}\}}\middle|\hat{\mathsf{M}}\right) && \text{by chain rule for entropy} \\
&= H\left(\mathbb{1}_{\{\hat{\mathsf{M}}\neq\mathsf{M}\}}\middle|\hat{\mathsf{M}}\right) + H\left(\mathsf{M}\middle|\hat{\mathsf{M}},\mathbb{1}_{\{\hat{\mathsf{M}}\neq\mathsf{M}\}}\right) && \text{by chain rule for entropy} \\
&\leq H\left(\mathbb{1}_{\{\hat{\mathsf{M}}\neq\mathsf{M}\}}\right) + H\left(\mathsf{M}\middle|\hat{\mathsf{M}},\mathbb{1}_{\{\hat{\mathsf{M}}\neq\mathsf{M}\}}\right) && \text{conditioning reduces entropy} \\
&= \hbar\left(P_{\mathsf{e}}^{(n)}\right) + P_{\mathsf{e}}^{(n)} H\left(\mathsf{M}\middle|\hat{\mathsf{M}},\mathbb{1}_{\{\hat{\mathsf{M}}\neq\mathsf{M}\}}=1\right) && \text{by } H\left(\mathsf{M}\middle|\hat{\mathsf{M}},\mathbb{1}_{\{\mathsf{M}\neq\hat{\mathsf{M}}\}}=0\right)=0 \text{ and} \\
& && H(\mathsf{A}|\mathsf{B}) = \sum_b \mathbf{P}(\mathsf{B}=b)\, H(\mathsf{A}|\mathsf{B}=b) \\
&\leq \hbar\left(P_{\mathsf{e}}^{(n)}\right) + P_{\mathsf{e}}^{(n)}\log(\lceil 2^{nR}\rceil - 1) && \text{because } H(\mathsf{Z}) \leq \log|\mathcal{Z}|
\end{aligned}$$

Proof of the Weak Converse Part II: Shannon capacity of product channels

$$
\begin{aligned}
I(\mathrm{X}_1^n;\mathrm{Y}_1^n) &\le C\left(p_{\mathrm{Y}_1^n|\mathrm{X}_1^n}\right) && \text{by the definition of } C(\cdot)\\
&= nC\left(p_{\mathrm{Y}|\mathrm{X}}\right) && \text{because } C\left(p_{\mathrm{Y}_1^n|\mathrm{X}_1^n}\right) = \sum\nolimits_t C\left(p_{\mathrm{Y}_t|\mathrm{X}_t}\right)
\end{aligned}
$$

Another proof of $I(\mathrm{X}_1^n;\mathrm{Y}_1^n) \le nC\left(p_{\mathrm{Y}|\mathrm{X}}\right)$.

$$
\begin{aligned}
I(\mathrm{X}_1^n;\mathrm{Y}_1^n) &= \sum\nolimits_{t=1}^n I\left(\mathrm{Y}_t;\mathrm{X}_1^n\middle|\mathrm{Y}_1^{t-1}\right) && \text{by the chain rule for M.I.}\\
&= \sum\nolimits_{t=1}^n H\left(\mathrm{Y}_t\middle|\mathrm{Y}_1^{t-1}\right) - H\left(\mathrm{Y}_t\middle|\mathrm{X}_1^n,\mathrm{Y}_1^{t-1}\right) && \text{by the definition of C.M.I.}\\
&= \sum\nolimits_{t=1}^n H\left(\mathrm{Y}_t\middle|\mathrm{Y}_1^{t-1}\right) - H\left(\mathrm{Y}_t|\mathrm{X}_t\right) && \text{by } p_{\mathrm{Y}_t|\mathrm{X}_1^n,\mathrm{Y}_1^{t-1}}\left(y_t|x_1^n,y_1^{t-1}\right) = p_{\mathrm{Y}_t|\mathrm{X}_t}(y_t|x_t)\\
&\le \sum\nolimits_{t=1}^n H(\mathrm{Y}_t) - H\left(\mathrm{Y}_t|\mathrm{X}_t\right) && \text{conditioning reduces entropy}\\
&= \sum\nolimits_{t=1}^n I(\mathrm{Y}_t;\mathrm{X}_t) && \text{by the definition of M.I.}\\
&\le \sum\nolimits_{t=1}^n C\left(p_{\mathrm{Y}|\mathrm{X}}\right) && \text{by the definition of } C(\cdot)
\end{aligned}
$$

Proof of the Weak Converse III

$$
\begin{aligned}
H\left(\mathrm{M}\middle|\hat{\mathrm{M}}\right) &\leq 1 + P_{\mathrm{e}}^{(n)} nR && \text{by Fano's inequality} && (*) \\
I(\mathrm{X}_1^n;\mathrm{Y}_1^n) &\leq nC\left(p_{\mathrm{Y}|\mathrm{X}}\right) && \text{Shannon capacity of product channels} && (**) \\
P_{\mathrm{e}}^{(n)} &\geq \frac{H\left(\mathrm{M}\middle|\hat{\mathrm{M}}\right)-1}{nR} && \text{by } (*) \\
&= \frac{H(\mathrm{M})-I\left(\mathrm{M};\hat{\mathrm{M}}\right)-1}{nR} && \text{by } I\left(\mathrm{M};\hat{\mathrm{M}}\right) = H(\mathrm{M}) - H\left(\mathrm{M}\middle|\hat{\mathrm{M}}\right) \\
&\geq \frac{nR-I\left(\mathrm{M};\hat{\mathrm{M}}\right)-1}{nR} && \text{because } H(\mathrm{M}) > nR \\
&\geq \frac{nR-I\left(\mathrm{X}_1^n;\mathrm{Y}_1^n\right)-1}{nR} && \text{by D.P.I. and } \mathrm{M} \leftrightarrow \mathrm{X}_1^n \leftrightarrow \mathrm{Y}_1^n \leftrightarrow \hat{\mathrm{M}} \\
&\geq \frac{nR-nC\left(p_{\mathrm{Y}|\mathrm{X}}\right)-1}{nR} && \text{by } (**) \\
&= \frac{R-C\left(p_{\mathrm{Y}|\mathrm{X}}\right)}{R} - \frac{1}{nR}
\end{aligned}
$$

Thus $P_{\mathrm{e}}^{(n)} > 0$ for large enough n provided that $R > C\left(p_{\mathrm{Y}|\mathrm{X}}\right)$.

Stronger Converse Results for Channel Coding Theorem

Definition

A channel $p_{Y|X}$ is said to satisfy *the strong converse* iff for any $R > C\left(p_{Y|X}\right)$ and $\epsilon > 0$, and for sufficiently large n, all $(\lceil 2^{Rn} \rceil, n)$ codes satisfy $P_{\mathbf{e}}^{(n)} \geq 1 - \epsilon$.

- Strong converse holds a channel $p_{Y|X}$ iff for all $\epsilon > 0$ there exists a γ_0 such that

$$\gamma \mathbf{P}\left(\frac{p_{Y|X}(Y|X)}{p_Y(Y)} > \gamma\right) \leq \epsilon \qquad \forall p_X,\ \gamma \geq \gamma_0$$

  where p_Y is the Y marginal of $p_{X,Y}$, see [Aug66, Aug78].
- For the source coding problem strong converse always holds: for all p_X with finite entropy if $R < H(X)$ then $\lim_{n\to\infty} P_{\mathbf{e}}^{(n)} = 1$
- For the channel coding problem we have following "medium converse": there exists a universal constant ϵ_0 such that if $R > C\left(p_{Y|X}\right)$ then $\liminf_{n\to\infty} P_{\mathbf{e}}^{(n)} \geq \epsilon_0$, [BC78].

Lemma (Strong Converse)

If $C\left(p_{Y|X}\right) < \infty$ and $\sigma < \infty$ where $\sigma = \sqrt{\sup_x \mathbf{E}\left[\left(\log \frac{p_{Y|X}(Y|x)}{p_{Y^\star}(Y)}\right)^2 \middle| X = x\right]}$ and $p_{Y^\star}$ is the unique Shannon center of $p_{Y|X}$ satisfying, then strong converse holds for $p_{Y|X}$.

Lemma (Strong Converse)

If $C\left(p_{Y|X}\right) < \infty$ *and* $\sigma < \infty$ *where* $\sigma = \sqrt{\sup_x \mathbf{E}\left[\left(\log \frac{p_{Y|X}(Y|x)}{p_{Y^\star}(Y)}\right)^2 \middle| X = x\right]}$ *and* $p_{Y^\star}$ *is the unique Shannon center of* $p_{Y|X}$*, satisfying, then strong converse holds for* $p_{Y|X}$*.*

Proof.

Let $Z_t := \log \frac{p_{Y|X}(Y_t|X_t)}{p_{Y^\star}(Y_t)}$ for all t and $S := \log \frac{p_{Y_1^n|X_1^n}(Y_1^n|X_1^n)}{p_{Y_1^n\star}(Y_1^n)}$.

$$S = \sum_{t=1}^{n} Z_t$$

$$\mathbf{E}\left[S|X_1^n\right] \leq nC \qquad \text{because } \mathbf{E}\left[Z_t|X_1^n\right] \leq C \tag{1}$$

$$\begin{aligned}
\mathbf{E}\left[(S - \mathbf{E}\left[S|X_1^n\right])^2|X_1^n\right] &= \sum_{t=1}^{n} \mathbf{E}\left[(Z_t - \mathbf{E}\left[Z_t|X_1^n\right])^2|X_1^n\right] && \text{because } Z_t\text{'s are cond. ind. given } X_1^n \\
&= \sum_{t=1}^{n} \mathbf{E}\left[(Z_t)^2|X_1^n\right] && \text{by } \mathbf{E}\left[(Z_t - \mathbf{E}\left[Z_t|X_1^n\right])^2|X_1^n\right] \leq \mathbf{E}\left[(Z_t)^2|X_1^n\right] \\
&\leq n\sigma^2 && (2)
\end{aligned}$$

$$\begin{aligned}
\mathbf{P}(S \geq nC + \gamma|X_1^n) &\leq \mathbf{P}(S \geq \mathbf{E}\left[S|X_1^n\right] + \gamma|X_1^n) && \text{by (1)} \\
&\leq \mathbf{P}(|S - \mathbf{E}\left[S|X_1^n\right]| \geq \gamma|X_1^n) && \text{for } \gamma \geq 0 \\
&\leq \frac{n\sigma^2}{\gamma^2} && \text{by Chebyshev's inequality and (2)}
\end{aligned}$$

Proof cont'd. $\mathbf{P}(\mathsf{S} \geq nC + \gamma | \mathsf{X}_1^n) \leq \frac{n\sigma^2}{\gamma^2}$ where $\mathsf{S} := \log \frac{p_{\mathsf{Y}_1^n|\mathsf{X}_1^n}(\mathsf{Y}_1^n|\mathsf{X}_1^n)}{p_{\mathsf{Y}_1^{n\star}}(\mathsf{Y}_1^n)}$ (3)

Let $\mathcal{A}^{(n)}(x_1^n) := \left\{ y_1^n : \log \frac{p_{\mathsf{Y}_1^n|\mathsf{X}_1^n}(y_1^n|x_1^n)}{p_{\mathsf{Y}_1^{n\star}}(y_1^n)} \leq n\left(C + \frac{\sigma}{n^{1/3}}\right) \right\}$

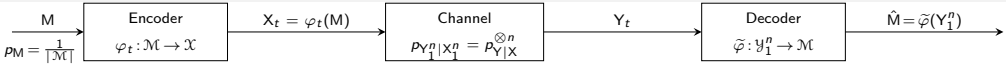
$$\mathbf{P}\left(\mathcal{E} \cap \mathcal{A}^{(n)}(\mathsf{X}_1^n) \middle| \mathsf{X}_1^n\right) \leq 2^{nC + n^{2/3}\sigma} \mathbf{P}(\mathsf{Y}_1^{n\star} \in \mathcal{E}) \quad \text{by def. of } \mathcal{A}^{(n)}(x_1^n) \quad (4)$$

$$\mathbf{P}\left(\mathsf{Y}_1^n \notin \mathcal{A}^{(n)}(\mathsf{X}_1^n) \middle| \mathsf{X}_1^n\right) \leq n^{-1/3} \quad \text{by (3) for } \gamma = \sigma n^{2/3} \quad (5)$$

$$\begin{aligned}
\mathbf{P}(\mathcal{E}|\mathsf{X}_1^n) &\leq \mathbf{P}\left(\mathsf{Y}_1^n \notin \mathcal{A}^{(n)}(\mathsf{X}_1^n) \middle| \mathsf{X}_1^n\right) + \mathbf{P}\left(\mathcal{E} \cap \mathcal{A}^{(n)}(\mathsf{X}_1^n) \middle| \mathsf{X}_1^n\right) \\
&\leq n^{-1/3} + 2^{nC + n^{2/3}\sigma} \mathbf{P}(\mathsf{Y}_1^{n\star} \in \mathcal{E}) && \text{by (4) and (5)} \quad (6)
\end{aligned}$$

$$\begin{aligned}
\mathbf{P}\left(\hat{\mathsf{M}} = \mathsf{M}\right) &= \frac{1}{\lceil 2^{nR} \rceil} \sum_m \mathbf{P}(\widetilde{\varphi}(\mathsf{Y}_1^n) = m | \mathsf{X}_1^n = \varphi(m)) \\
&\leq \frac{1}{\lceil 2^{nR} \rceil} \sum_m \left(n^{-1/3} + 2^{nC + n^{2/3}\sigma} \mathbf{P}(\widetilde{\varphi}(\mathsf{Y}_1^{n\star}) = m) \right) && \text{by (6)} \\
&= n^{-1/3} + \frac{1}{\lceil 2^{nR} \rceil} 2^{nC + n^{2/3}\sigma} && \text{by } \sum_m \mathbf{P}(\widetilde{\varphi}(\mathsf{Y}_1^{n\star}) = m) = 1 \\
&\leq n^{-1/3} + 2^{-n(R - C - \frac{\sigma}{n^{1/3}})}
\end{aligned}$$

Memoryless Channels with and without Feedback

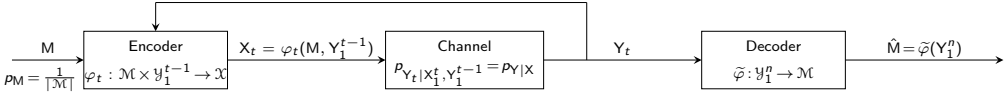


DMC $p_{Y_1^n|X_1^n}(y_1^n|x_1^n) = \prod_{t=1}^n p_{Y|X}(y_t|x_t) \qquad \forall x_1^n, y_1^n$

Encoding function $\varphi = [\varphi_1, \ldots, \varphi_n]$ where $\varphi_t : \mathcal{M} \to \mathcal{X}$

Decoding function $\widetilde{\varphi} : \mathcal{Y}_1^n \to \mathcal{M}$.

$$P_e^m = \sum_{y_1^n : \widetilde{\varphi}(y_1^n) \neq m} \prod_{t=1}^n p_{Y|X}(y_t|\varphi_t(m)).$$



DMC $p_{Y_t|X_1^t, Y_1^{t-1}}\left(y_t|x_1^t, y_1^{t-1}\right) = p_{Y|X}(y_t|x_t) \qquad \forall x_1^t, y_1^t, t$

Encoding function $\varphi = [\varphi_1, \ldots, \varphi_n]$ where $\varphi_t : \mathcal{M} \times \mathcal{Y}_1^{t-1} \to \mathcal{X}$

Decoding function $\widetilde{\varphi} : \mathcal{Y}_1^n \to \mathcal{M}$.

$$P_e^m = \sum_{y_1^n : \widetilde{\varphi}(y_1^n) \neq m} \prod_{t=1}^n p_{Y|X}\left(y_t|\varphi_t(m, y_1^{t-1})\right)$$

Does maximum achievable rate increase with feedback?

Proof of the Weak Converse For Memoryless Channels With Feedback I

$$
\begin{aligned}
I(\mathrm{M};\mathrm{Y}_1^n) &= \sum_{t=1}^{n} I\left(\mathrm{M};\mathrm{Y}_t \middle| \mathrm{Y}_1^{t-1}\right) && \text{by the chain rule for M.I.}\\
&= \sum_{t=1}^{n} H\left(\mathrm{Y}_t \middle| \mathrm{Y}_1^{t-1}\right) - H\left(\mathrm{Y}_t \middle| \mathrm{M},\mathrm{Y}_1^{t-1}\right) && \text{by the definition of C.M.I.}\\
&= \sum_{t=1}^{n} H\left(\mathrm{Y}_t \middle| \mathrm{Y}_1^{t-1}\right) - H\left(\mathrm{Y}_t \middle| \mathrm{X}_t\right) && \text{by } p_{\mathrm{Y}_t|\mathrm{M},\mathrm{Y}_1^{t-1}}\left(y_t \middle| m, y_1^{t-1}\right) = p_{\mathrm{Y}_t|\mathrm{X}_t}\left(y_t \middle| \varphi_t(m, y_1^{t-1})\right)\\
&\leq \sum_{t=1}^{n} H(\mathrm{Y}_t) - H\left(\mathrm{Y}_t \middle| \mathrm{X}_t\right) && \text{conditioning reduces entropy}\\
&= \sum_{t=1}^{n} I(\mathrm{X}_t;\mathrm{Y}_t) && \text{by the definition of M.I.}\\
&\leq \sum_{t=1}^{n} C\left(p_{\mathrm{Y}_t|\mathrm{X}_t}\right) && \text{by the definition of } C(\cdot)\\
&= nC\left(p_{\mathrm{Y}|\mathrm{X}}\right) && \text{because } p_{\mathrm{Y}_t|\mathrm{X}_t} = p_{\mathrm{Y}|\mathrm{X}}
\end{aligned}
$$

Remark

$\varphi(m) = [\varphi_1(m), \ldots, \varphi_n(m)]$ for each $m \in \mathcal{M}$ is an input letter for product channel with feedback. (x_1^n is an input letter for product channel.) The Shannon capacity of the product channel with feedback is equal to the Shannon capacity of the product channel (without feedback).

Proof of the Weak Converse For Memoryless Channels With Feedback II

$$
\begin{aligned}
H\left(\mathrm{M}\middle|\hat{\mathrm{M}}\right) &\leq 1 + P_{\mathrm{e}}^{(n)} nR && \text{by Fano's inequality} && (*) \\
I(\mathrm{M}; \mathrm{Y}_1^n) &\leq nC\left(p_{\mathrm{Y}|\mathrm{X}}\right) && \text{(Shannon capacity of the product channels with feedback)} && (**) \\
P_{\mathrm{e}}^{(n)} &\geq \frac{H\left(\mathrm{M}\middle|\hat{\mathrm{M}}\right)-1}{nR} && \text{by } (*) \\
&= \frac{H(\mathrm{M})-I\left(\mathrm{M};\hat{\mathrm{M}}\right)-1}{nR} && \text{by } I\left(\mathrm{M};\hat{\mathrm{M}}\right) = H(\mathrm{M}) - H\left(\mathrm{M}\middle|\hat{\mathrm{M}}\right) \\
&\geq \frac{nR-I\left(\mathrm{M};\hat{\mathrm{M}}\right)-1}{nR} && \text{because } H(\mathrm{M}) = \log\lceil 2^{nR}\rceil \geq nR \\
&\geq \frac{nR-I\left(\mathrm{M};\mathrm{Y}_1^n\right)-1}{nR} && \text{by D.P.I. and } \mathrm{M} \leftrightarrow \mathrm{Y}_1^n \leftrightarrow \hat{\mathrm{M}} \\
&\geq \frac{nR-nC\left(p_{\mathrm{Y}|\mathrm{X}}\right)-1}{nR} && \text{by } (**) \\
&= \frac{R-C\left(p_{\mathrm{Y}|\mathrm{X}}\right)}{R} - \frac{1}{nR}
\end{aligned}
$$

Thus $P_{\mathrm{e}}^{(n)} > 0$ for large enough n provided that $R > C\left(p_{\mathrm{Y}|\mathrm{X}}\right)$.

Strong Converse For Memoryless Channels With Feedback

Definition

A channel $p_{\mathrm{Y}|\mathrm{X}}$ is said to satisfy *the strong converse* iff for any $R > C\left(p_{\mathrm{Y}|\mathrm{X}}\right)$ and $\epsilon > 0$, and for sufficiently large n, all $(\lceil 2^{Rn} \rceil, n)$ codes satisfy $P_{\mathbf{e}}^{(n)} \geq 1 - \epsilon$.

Lemma (Strong Converse)

If $C\left(p_{\mathrm{Y}|\mathrm{X}}\right) < \infty$ *and* $\sigma < \infty$ *where* $\sigma = \sqrt{\sup_x \mathbf{E}\left[\left(\log \frac{p_{\mathrm{Y}|\mathrm{X}}(\mathrm{Y}|x)}{p_{\mathrm{Y}^\star}(\mathrm{Y})}\right)^2 \middle| \mathrm{X} = x\right]}$ *and* $p_{\mathrm{Y}^\star}$ *is the unique Shannon center of* $p_{\mathrm{Y}|\mathrm{X}}$ *satisfying, then strong converse holds product channels with feedback obtained from* $p_{\mathrm{Y}|\mathrm{X}}$.

Given M, channel outputs $\mathrm{Y}_1, \ldots, \mathrm{Y}_n$ are jointly independent for memoryless channels, but this not the case for memoryless channels with feedback.

In the proof of the strong converse for memoryless channels, however, we have used the independence only to ensure the uncorrelatedness of $\mathrm{Z}_1, \ldots, \mathrm{Z}_n$ for $\mathrm{Z}_t := \log \frac{p_{\mathrm{Y}|\mathrm{X}}(\mathrm{Y}_t|\mathrm{X}_t)}{p_{\mathrm{Y}^\star}(\mathrm{Y}_t)}$.

Proof of the Strong Converse For Memoryless Channels With Feedback I

Let $Y_0 := M$, $Z_t := \log \frac{p_{Y|X}\left(Y_t|\varphi_t(Y_0^{t-1})\right)}{p_{Y^\star}(Y_t)}$ and $\widetilde{Z}_t := Z_t - \mathbf{E}\left[Z_t \middle| Y_0^{t-1}\right]$ $\forall t$ and $S := \log \frac{p_{Y_1^n|Y_0}(Y_1^n|Y_0)}{p_{Y_1^{n\star}}(Y_1^n)}$.

$$\mathbf{E}\left[\widetilde{Z}_t \widetilde{Z}_{t-k} \middle| Y_0\right] = 0 \qquad \text{by } \mathbf{E}\left[\widetilde{Z}_t \middle| Y_0^{t-1}\right] = 0 \ \& \ \mathbf{E}\left[\widetilde{Z}_{t-k} \middle| Y_0^{t-1}\right] = \widetilde{Z}_{t-k} \tag{7}$$

$$\mathbf{E}\left[\left(\sum_{t=1}^n \widetilde{Z}_t\right)^2 \middle| Y_0\right] = \sum_{t=1}^n \mathbf{E}\left[(\widetilde{Z}_t)^2 \middle| Y_0\right] \qquad \text{by (7)}$$

$$\leq n\sigma^2 \qquad \text{by } \mathbf{E}\left[(\widetilde{Z}_t)^2 \middle| Y_0\right] = \mathbf{E}\left[\mathbf{E}\left[(\widetilde{Z}_t)^2 \middle| Y_0^{t-1}\right] \middle| Y_0\right] \tag{8}$$

$$\mathbf{P}\left(\left|\sum_{t=1}^n \widetilde{Z}_t\right| \geq \gamma \middle| Y_0\right) \leq \frac{n\sigma^2}{\gamma^2} \qquad \text{for } \gamma > 0 \text{ by Chebyshev's inequaity \& (8)} \tag{9}$$

$$S = \sum_{t=1}^n Z_t \qquad \text{by } p_{Y|Y_0^{t-1}}\left(Y_t|Y_0^{t-1}\right) = p_{Y|X}\left(Y_t|\varphi_t(Y_0^{t-1})\right) \tag{10}$$

$$\mathbf{E}\left[Z_t \middle| Y_0^{t-1}\right] \leq C \qquad \text{by } S\left(p_{Y|X}\right) = C\left(p_{Y|X}\right) \tag{11}$$

$$S - nC \leq \sum_{t=1}^n \widetilde{Z}_t \qquad \text{by (10) and (11)} \tag{12}$$

$$\mathbf{P}(S \geq nC + \gamma | Y_0) \leq \frac{n\sigma^2}{\gamma^2} \qquad \text{by (9) and (12)}$$

Proof of the Strong Converse For Memoryless Channels With Feedback II

$$\mathbf{P}(\mathsf{S}(\mathsf{M}) \geq nC + \gamma | \mathsf{M}) \leq \frac{n\sigma^2}{\gamma^2} \qquad \text{where } \mathsf{S}(m) := \log \frac{p_{\mathsf{Y}_1^n|\mathsf{M}}(\mathsf{Y}_1^n|m)}{p_{\mathsf{Y}_1^{n\star}}(\mathsf{Y}_1^n)} \tag{13}$$

Let $\mathcal{A}^{(n)}(m) := \{y_1^n : \mathsf{S}(m) \leq nC + \sigma n^{2/3}\}$

$$\mathbf{P}\left(\mathcal{E} \cap \mathcal{A}^{(n)}(\mathsf{M}) \middle| \mathsf{M}\right) \leq 2^{nC+n^{2/3}\sigma} \mathbf{P}(\mathsf{Y}_1^{n\star} \in \mathcal{E}) \qquad \text{by def. of } \mathcal{A}^{(n)}(m) \tag{14}$$

$$\mathbf{P}\left(\mathsf{Y}_1^n \notin \mathcal{A}^{(n)}(\mathsf{M}) \middle| \mathsf{M}\right) \leq n^{-1/3} \qquad \text{by (13) for } \gamma = \sigma n^{2/3} \tag{15}$$

$$\begin{aligned}
\mathbf{P}(\mathcal{E}|\mathsf{M}) &\leq \mathbf{P}\left(\mathsf{Y}_1^n \notin \mathcal{A}^{(n)}(\mathsf{M}) \middle| \mathsf{M}\right) + \mathbf{P}\left(\mathcal{E} \cap \mathcal{A}^{(n)}(\mathsf{M}) \middle| \mathsf{M}\right) \\
&\leq n^{-1/3} + 2^{nC+n^{2/3}\sigma} \mathbf{P}(\mathsf{Y}_1^{n\star} \in \mathcal{E}) && \text{by (14) and (15)} \qquad (16)
\end{aligned}$$

$$\begin{aligned}
\mathbf{P}\left(\hat{\mathsf{M}} = \mathsf{M}\right) &= \frac{1}{\lceil 2^{nR} \rceil} \sum_m \mathbf{P}(\widetilde{\varphi}(\mathsf{Y}_1^n) = m | \mathsf{M} = \varphi(m)) \\
&\leq \frac{1}{\lceil 2^{nR} \rceil} \sum_m \left(n^{-1/3} + 2^{nC+n^{2/3}\sigma} \mathbf{P}(\widetilde{\varphi}(\mathsf{Y}_1^{n\star}) = m)\right) && \text{by (16)} \\
&= n^{-1/3} + \frac{1}{\lceil 2^{nR} \rceil} 2^{nC+n^{2/3}\sigma} && \text{by } \sum_m \mathbf{P}(\widetilde{\varphi}(\mathsf{Y}_1^{n\star}) = m) = 1 \\
&\leq n^{-1/3} + 2^{-n(R-C)+\sigma n^{2/3}}
\end{aligned}$$

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